Elangoraj Thiruppandiaraj

AI & Data Product Leader | Generative AI, Enterprise Platforms & Financial Services

London, UK | +44 7586073892 | elangorj@gmail.com | LinkedIn | Github | Medium | Portfolio

Professional Summary

AI and Data Product Leader with 10 years of experience delivering enterprise AI, analytics, and data-driven platforms across financial services, insurance, and HR technology. Experienced in leading cross-functional initiatives spanning product strategy, AI solution architecture, stakeholder engagement, and scalable platform delivery. Proven track record of translating complex business challenges into enterprise AI products, driving operational efficiency, automation, and strategic adoption of Generative AI technologies within regulated environments.

Key Skills & Technical Stack

- **Product, Strategy & Leadership**: AI Product Strategy, Enterprise AI Platforms, Stakeholder Management, Cross-Functional Leadership, AI Roadmaps, Executive Communication, Delivery Governance, Agile Delivery, Technical Discovery, AI Capability Enablement, Consulting & Advisory, Team Leadership, Mentoring, Business-Technology Alignment.
- **Generative AI & Enterprise AI Solutions**: Generative AI, Retrieval-Augmented Generation (RAG), Agentic Workflows, Multi-Agent Systems, Prompt Engineering, AI Assistants, Natural Language Interfaces, Document Intelligence, Structured Output Generation, Semantic Search, LLM Evaluation, AI Workflow Automation.
- **Data Science & Machine Learning**: Predictive Modelling, NLP, Recommendation Systems, Anomaly Detection, Computer Vision, OCR, Statistical Analysis, Time Series Forecasting, Supervised & Unsupervised Learning, Transformer Models, Model Evaluation, A/B Testing.
- **AI Engineering & Platform Technologies**: Python, SQL, Spark, MLflow, Docker, GitHub, CI/CD, MLOps, LLMOps, Model Deployment, API Integration, Vector Databases, Embeddings, Streamlit, Databricks, Azure DevOps.
- **Financial Services & Governance**: Financial Services, Insurance, Underwriting, Risk Assessment, Compliance, Responsible AI, AI Governance, Data Governance, Privacy-by-Design, Regulated Enterprise Environments.
- **Cloud & Enterprise Technologies**: Azure ML, Azure AI Foundry, Azure Databricks, Azure App Services, Azure DevOps, AWS (S3, EMR, SageMaker, Redshift), GCP Vertex AI, BigQuery, Snowflake, Power BI.
- **Cloud Platforms:** Azure ML, Azure AI Foundry, Azure Databricks, Azure App Services, Azure DevOps, GCP Vertex AI, BigQuery, AWS S3, EMR, SageMaker, Redshift.

Work History

Data Scientist (AI Solution Lead) | Just Group Plc, London, UK | July 2024- Present

- Lead enterprise AI product initiatives across underwriting, risk, and retail functions, owning discovery, stakeholder alignment, platform direction, and delivery of scalable AI-enabled workflows across regulated business environments.
- **Document Intelligence & RAG Platform:** Designed and deployed a secure document understanding system used by medical underwriters to analyse complex reports and financial documents, automating highly-structured decision-making workflows similar to **credit and risk rating processes**. Built a multimodal pipeline combining OCR-based extraction from scanned images and handwritten notes, vector embeddings for semantic indexing, similarity search across document chunks, and re-ranking strategies to retrieve high-confidence answers. The platform processes structured and unstructured medical and financial data and is being expanded to support additional business workflows.

- **Agentic AI Platform:** Architected and delivered a multi-agent Generative AI system enabling asynchronous orchestration of multiple GPT-family models to generate structured outputs and automatically populate template-based documents. The system processes transcripts, PDFs, and structured inputs to produce refined reports and operational notes, saving hundreds of hours of manual effort each month across multiple business teams.
- Led discovery workshops and consulting-style engagements with underwriting, risk, compliance, and business stakeholders to identify AI opportunities, assess feasibility, define roadmaps, and translate domain expertise into machine-readable rules, checks, and evaluation criteria.
- Defined platform roadmap, prioritised AI use cases, and drove enterprise adoption of internal AI solutions across multiple business units.
- **Anomaly Detection for Financial Data:** Developed a similarity-search–based anomaly detection approach using embedding representations to identify irregular patterns within quote and underwriting datasets. The system detects subtle behavioural deviations and data inconsistencies that traditional rule-based checks fail to capture, improving early detection of potential risk or fraud-like patterns.
- Partner with business, IT, and compliance teams to ensure AI solutions align with regulatory and governance frameworks; regularly present AI architecture decisions, roadmap progress, and ROI metrics to the **Group Executive Committee** to secure continued investment in AI capabilities.
- Founded and lead the **AI4U initiative and AI Centre of Excellence activities**, driving AI adoption through use-case discovery, prompt engineering workshops, capability sessions, and internal consulting, reaching **200+ active participants and over 40% of the organisation including executive leadership**.
- Represent Just Group externally as an AI speaker at industry conferences including **ai:NOW 2025 and IRM Europe 2025**, presenting on Responsible AI, AI in Financial Services, and Agentic AI systems.

AI Product Manager | Mercer, London, UK | Dec 2021- June 2024

- Owned the roadmap and end-to-end delivery of an AI-driven internal product, operating at the intersection of product, data science, and engineering. Led discovery workshops with business stakeholders to define requirements and translate them into scalable ML solutions.
- Managed and mentored a cross-functional team of two AI/Data Engineers, acting as the bridge between business stakeholders and technical teams to ensure alignment between product strategy, technical feasibility, and delivery outcomes.
- Designed and implemented a multilingual transformer-based semantic matching system using encoder-based architectures and contextual embeddings from Hugging Face models, improving matching accuracy by **73%** and reducing operational costs by **33%.**
- Led experimentation and optimisation of transformer models using **Parameter Efficient Fine-Tuning (PEFT)**, achieving a **15% performance improvement** over base models.
- Established **MLOps practices** including model tracking, versioning, and reproducibility using MLflow and cloud infrastructure, while exploring Generative AI approaches such as **RAG, RLHF, and LangChain** to improve contextual intelligence across products.

Research Intern (AI) | Intelligent AI, Exeter, UK | April – May 2021

- Developed a satellite image scene-classification model to detect ten land-use categories (e.g., rivers, residences, forests) using Python, PyTorch/TensorFlow, and Docker.

Module lead | LTIMindtree, Chennai, India | Jan 2016 - Oct 2021

- Led the development of a deep neural network–based recommendation engine for Unilever using Apache Spark and Python, supported by early MLOps practices for experiment tracking, model evaluation, and deployment workflows. Improving product placement accuracy to **80%** and ensuring optimal SKU distribution across retail outlets.

- Built and maintained **data engineering pipelines using SQL, Python, and Apache Spark**, processing large retail datasets for machine learning training, analytics, and operational reporting.
- Developed **Power BI dashboards and reporting solutions** for business stakeholders, translating complex data insights into actionable recommendations for supply chain and sales teams.
- Delivered **internal training sessions on machine learning and data science**, contributing to wider adoption of ML/DL practices across engineering teams as part of the Data Science Centre of Excellence.
- Worked onsite at client locations for over **one year**, collaborating directly with business stakeholders to gather requirements, perform data processing, and build reporting and analytics solutions aligned with operational needs.

Publication

Remote Sensing Single Image Super-Resolution Benchmarking with Transfer Learning Algorithms, 2023, Publisher: IEEE, Co-Author: Saptarshi Das.

Education

MSc Data Science with Artificial Intelligence (Distinction) University of Exeter, UK | 2020-2021
Bachelors in Electronics and Communication St. Joseph's College of Engineering, India | 2011-2015

Certifications

- Google Project Management, Generative AI with Large Language Model
- Technical Team Leadership (**University of Oxford),** Deep learning Nanodegree

LEADERSHIP & COMMUNITY

- **Vice President Education, Toastmasters International (City of London)** – Lead club education, meetings, support member development, and drive engagement. Awarded the Future Forward Award for advancing member growth and club performance.
- **AI Mentor & Career Coach** – Support professionals transitioning into AI and tech through mentoring, external coaching, and Fiverr, focusing on learning paths, portfolios, and job.

Languages: Tamil, English